

Joshua Mark

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EDUCATION

UNC Chapel Hill

Expected: May 2027

B.S. Geospatial Data Science, B.S. Information Science, Minor Music

Chapel Hill, NC

- **Cumulative GPA:** 3.991/4.0 | **Dean's List:** all semesters.
- **Relevant Coursework:** Applied Spatial Data Science, GIS Programming, Quantitative Methods in Geography, Cartography & Geovisualization, Intro to Remote Sensing, Mathematical Tools for Data Science.

TECHNICAL SKILLS

GIS & Spatial Analysis: ArcGIS Pro, ArcGIS Online, QGIS, GeoDa, Google Earth Engine.

Programming & Data Analysis: Python (ArcPy, pandas); R (sf, tmap, tidyverse, tidycensus, terra); SQL.

Spatial & Statistical Methods: Exploratory data analysis, Regression (OLS + spatial) modeling, hypothesis testing.

Visualization & Development: Tableau, HTML, CSS, PHP, Git/GitHub.

PROJECTS

[North Carolina Triangle Walkability Analysis](#)

- Developed a four-script pipeline in R, integrating EPA Walkability Index with the US Housing and Urban Development's Low-to-Moderate Income (LMI) dataset to find a relationship between walkability and income within North Carolina Triangle.
- Conducted OLS regression, then applied a Spatial Lag and Spatial Autoregressive Moving Average (SARMA) model and evaluated suitability using spatial autocorrelation diagnostics and Lagrange Multiplier (LM) tests.
- Produced a bivariate choropleth map revealing significant patterns contrary to the initial research hypothesis.

[Data Centers and Electricity Tool](#)

- Acquired, cleaned, and integrated datasets containing Zip Code-level electricity rates and crowdsourced data center locations.
- Built ArcPy geoprocessing tool calculating proximity and electricity cost metrics within user-defined Zip Codes.
- Performed a ten-year time-series analysis of electricity rate changes and incorporating findings into comparative data center assessments.

[Residential Access to Grocery Stores](#)

- Conducted network-based accessibility analysis using QGIS Open Route Service (ORS) to identify areas within a five-minute drive or twenty-minute walk from grocery stores.
- Integrated Chapel Hill jurisdictional boundary neighborhood datasets to accurately assess residential access.
- Designed a professional cartographic map highlighting areas with limited access to grocery stores.

[Comparative Segregation Analysis of Detroit and Tampa](#)

- Investigated spatial patterns of inequality using the Index of Concentrations at the Extremes (ICE) across Detroit and Tampa.
- Performed an exploratory spatial analysis, including descriptive statistics, hotspot analysis, and cluster detection.
- Developed multivariate spatial regression models to examine the impact of historical redlining, home values, and environmental health indicators. Interpreted spatial nonstationarity and contextualized findings using local historical and socioeconomic conditions.

RESEARCH EXPERIENCE

UNC Chapel Hill, Human-AI Interactions in Knowledge Work

2024 – 2025

Undergrad Research Assistant

Chapel Hill, NC

- Supported participant recruitment and screening for a multi-cohort study examining human-AI interactions.
- Managed Qualtrics survey entries and participant communication for 54 study participants over 3 cohorts.
- Conducted semi-structured interviews with 10 participants and synthesized qualitative findings to support ongoing research study.